

Incident Management Executive Overview

Incident volume, resolution performance, SLA compliance, and operational trends

Created by *Jake Sheeler*

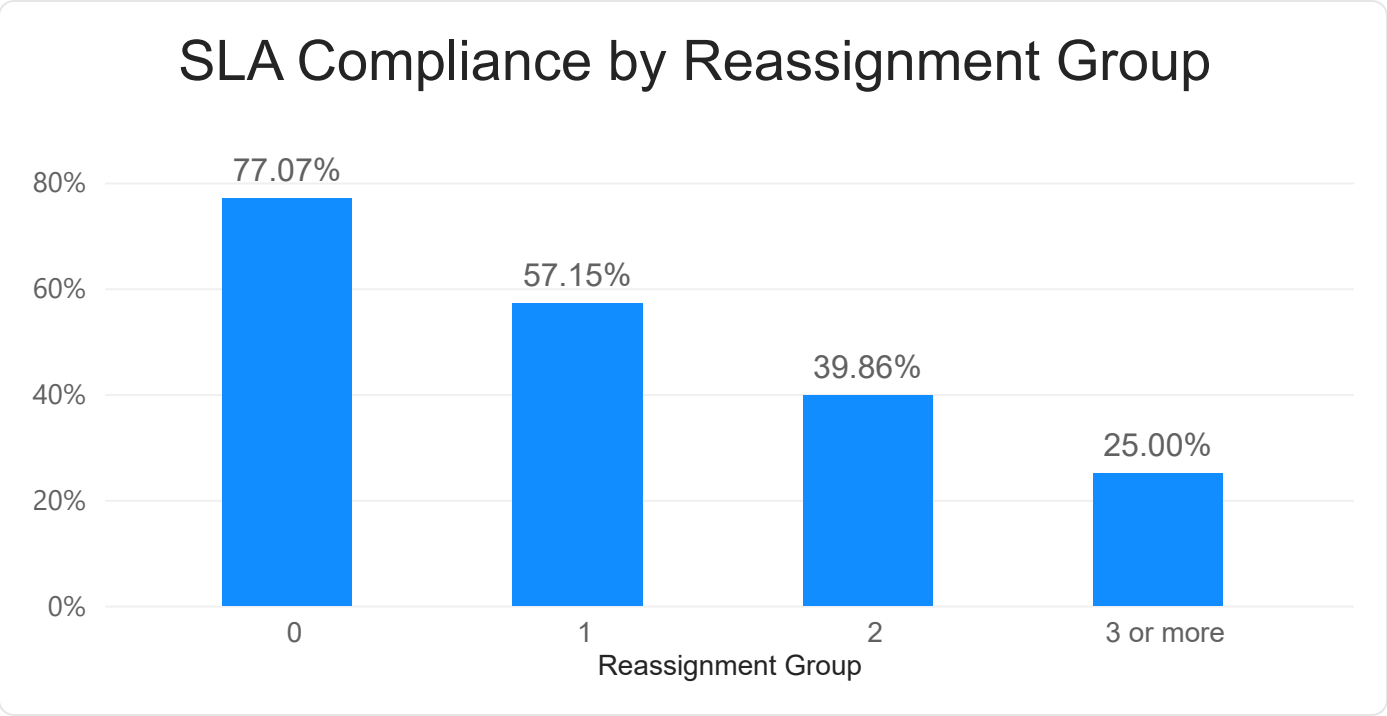
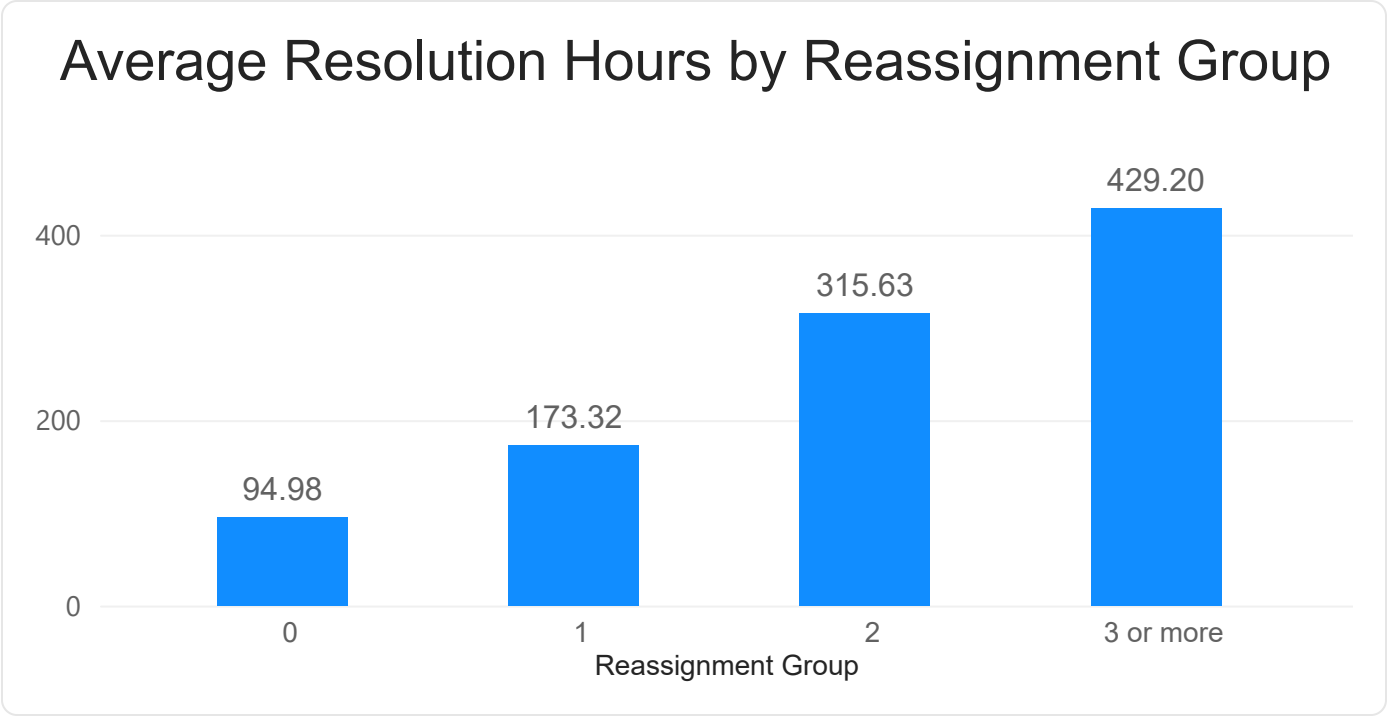
Total Incidents
24,918

Avg Resolution Hours
178.17

SLA Compliance %
63.42%

Reassigned Incidents
11,369

Reopened Incidents
275



Priority

All

Opened Month

All

Priority and Month filters apply to summary metrics only.
Charts display overall performance benchmarks.



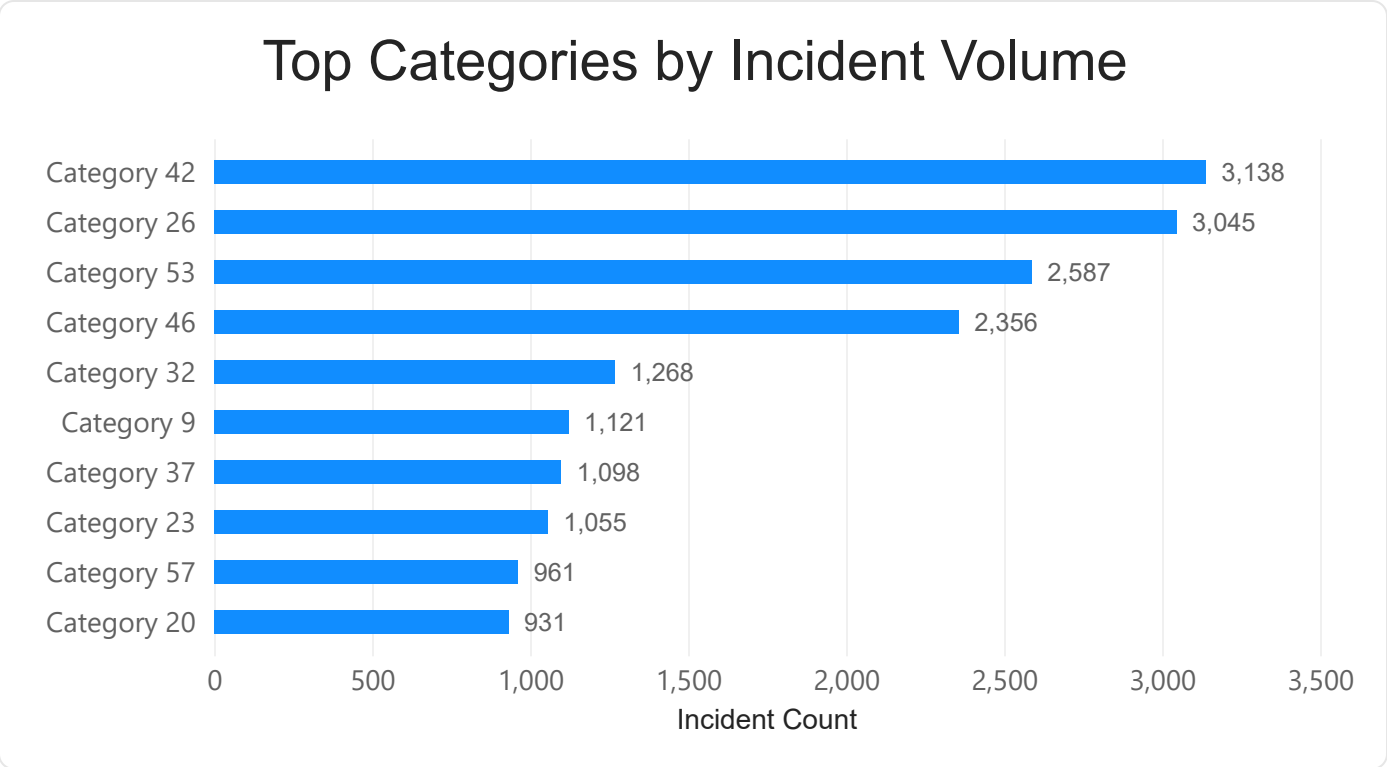
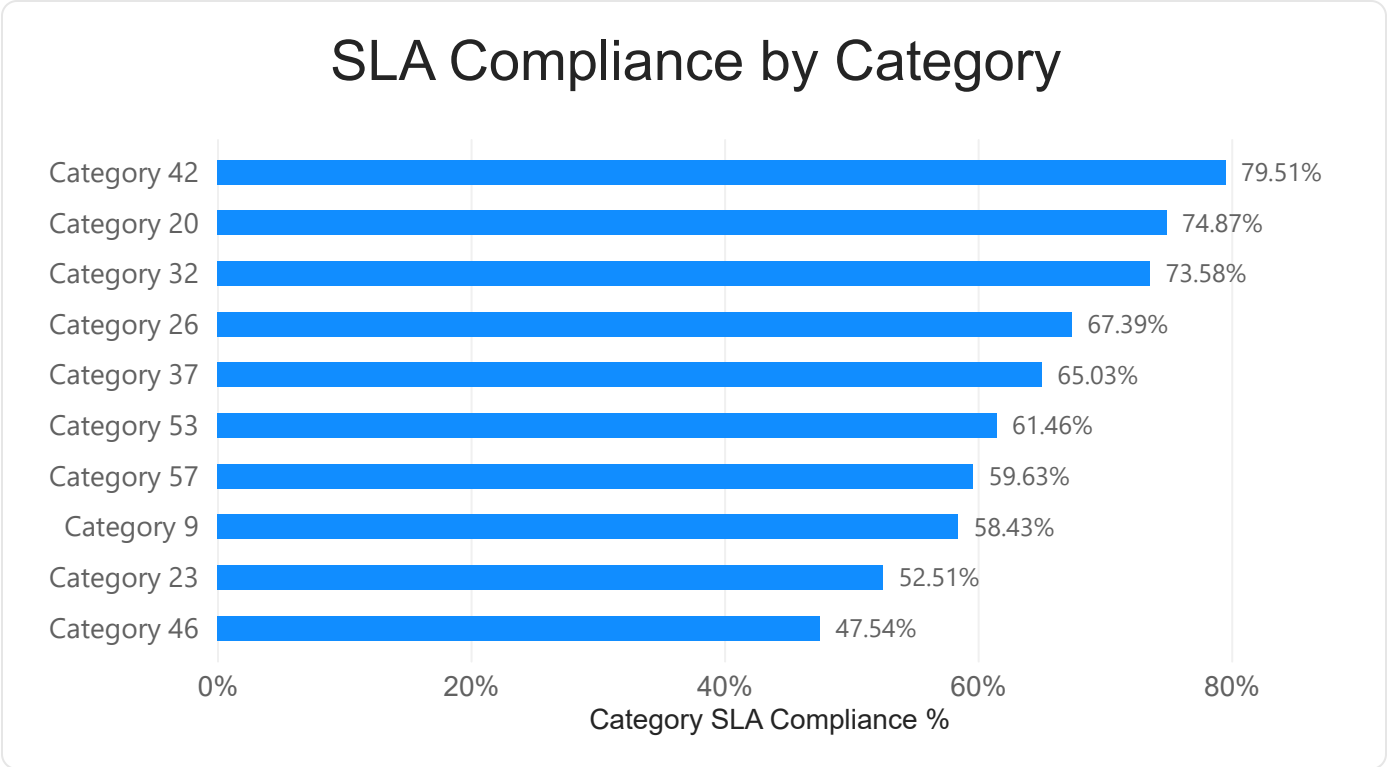
Key Findings

Incidents with three or more reassignments averaged **429.20 resolution hours**, compared with **94.98 hours** for incidents with no reassignments.

SLA compliance decreased from **77.07%** with no reassignments to **25.00%** with three or more reassignments.

Category 46 recorded **2,356 incidents** with **47.54% SLA compliance**, making it a high volume performance concern.

Category 42 handled the highest incident volume among the displayed categories while maintaining **79.51% SLA compliance**.



Reassignment Analysis

Examining how ticket handoffs relate to resolution time, SLA compliance, and reopen rates

No Reassignment Resolution Hours
94.98

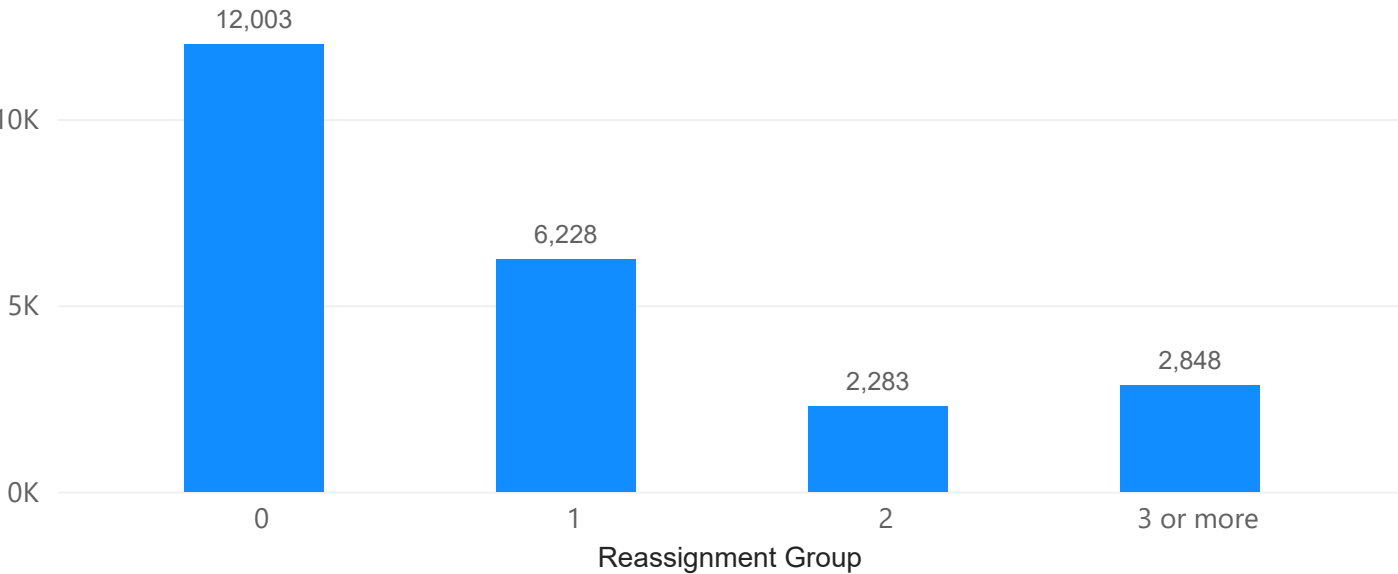
3+ Reassignment Resolution Hours
429.20

No Reassignment SLA
77.07%

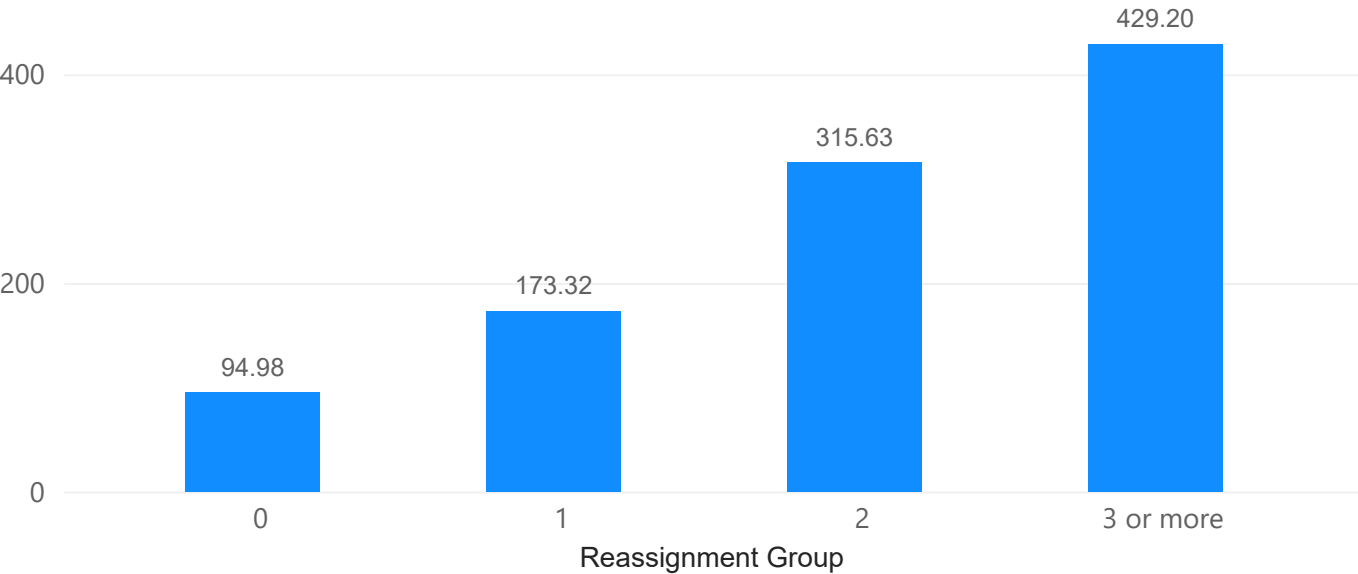
3+ Reassignment SLA
25.00%

Reassignment Rate
45.63%

Incident Volume by Reassignment Group



Average Resolution Hours by Reassignment Group



Key Findings

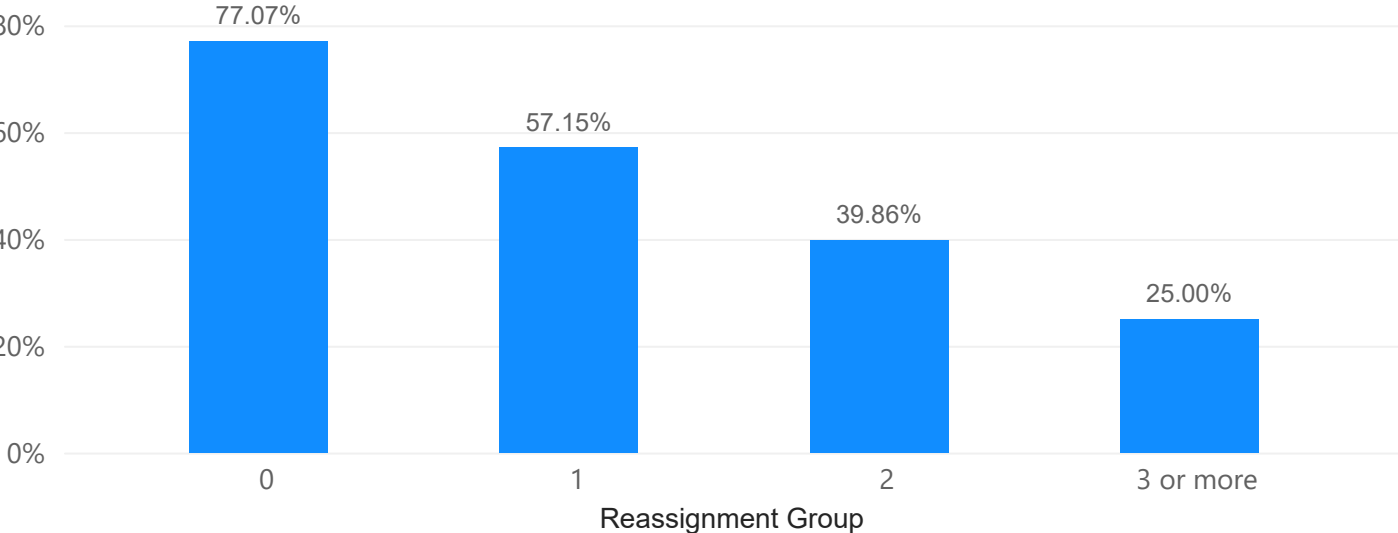
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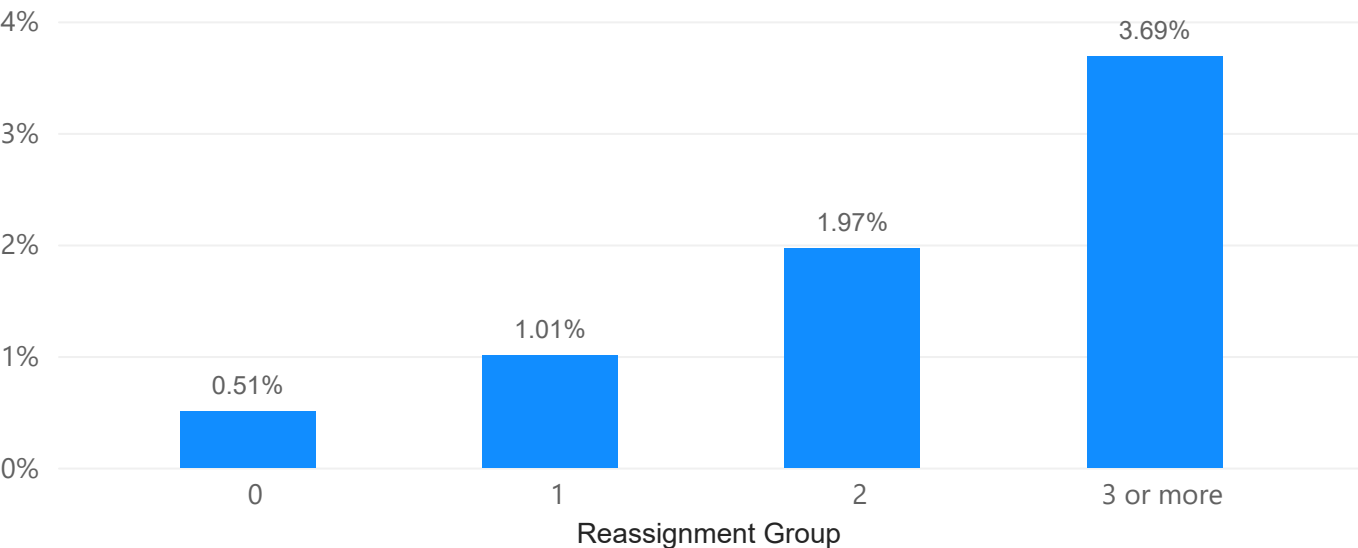
Reopen rates increased from **0.51%** for incidents with no reassignments to **3.69%** for incidents with three or more reassignments.

45.63% of all incidents experienced at least one reassignment, representing **11,369 of 24,918 incidents**.

SLA Compliance % by Reassignment Group



Reopen Rate by Reassignment Group



Category Performance

Comparing incident volume, resolution time, SLA compliance, and reopen rates across service categories

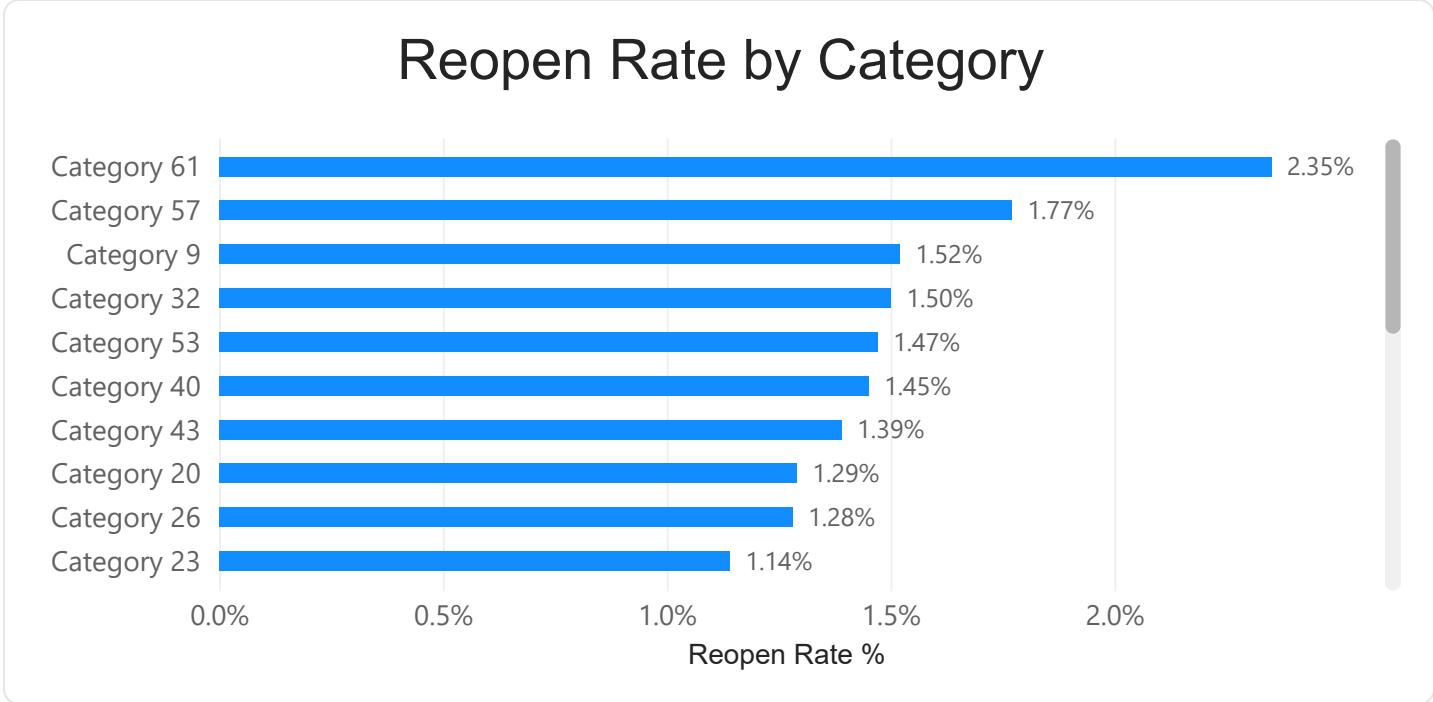
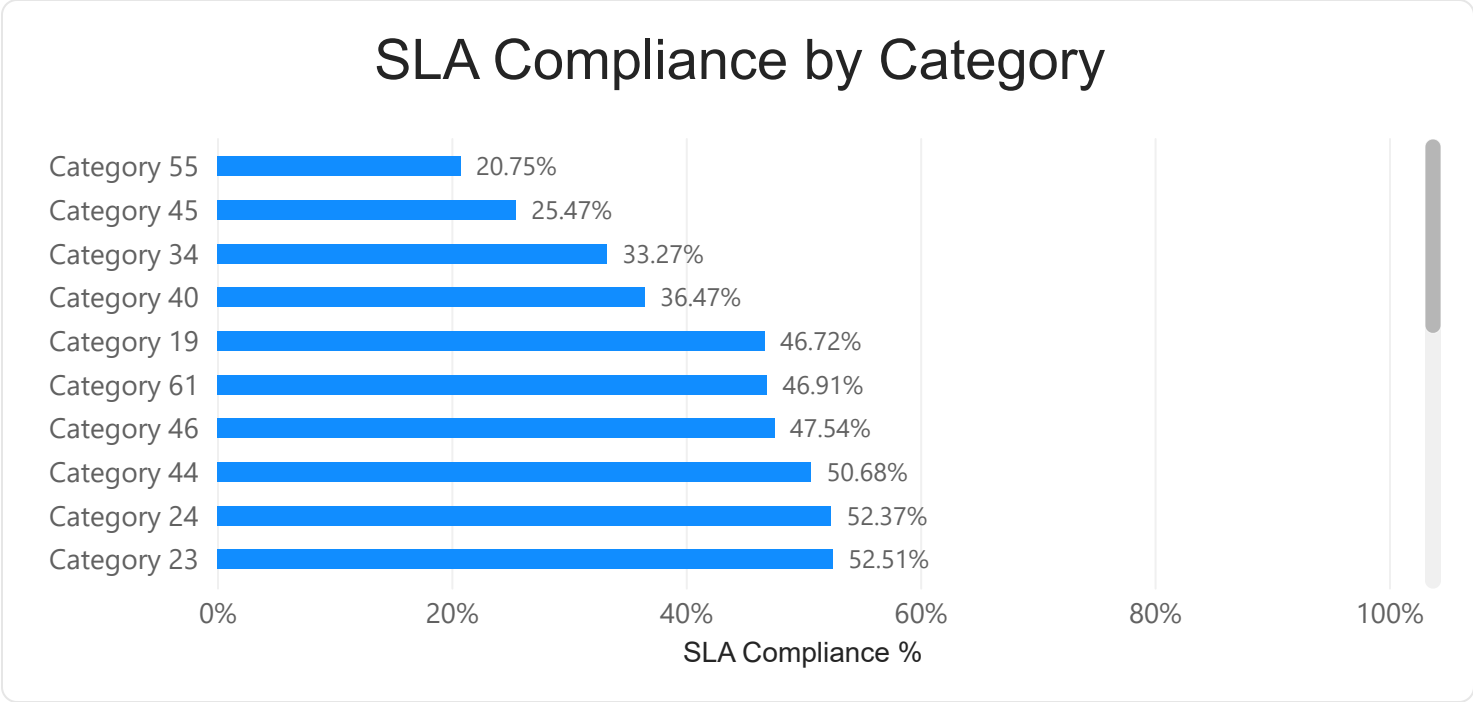
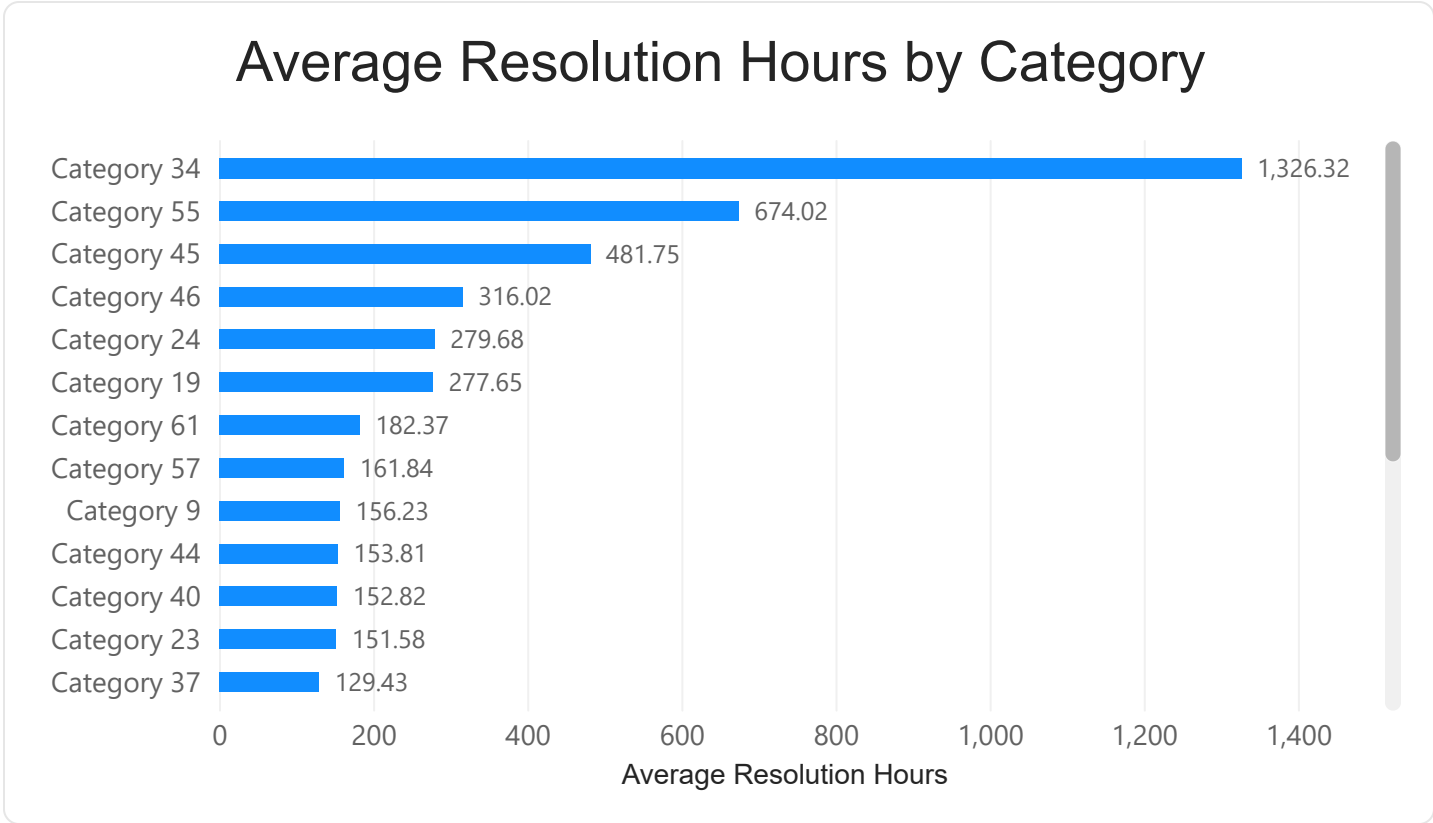
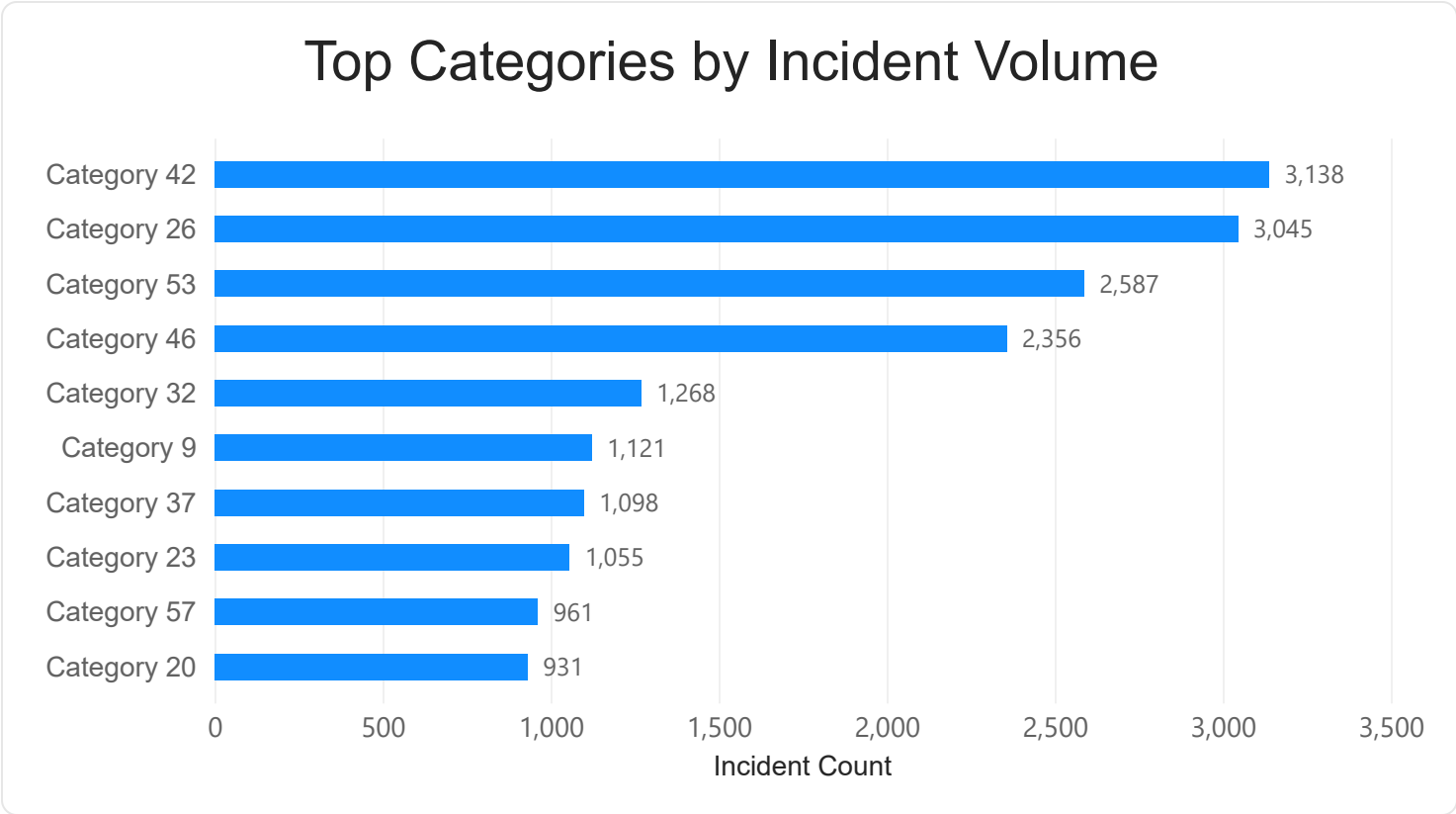
Highest Volume Category
Category 42

Highest Incident Volume
3,138

Highest Avg Resolution Category
Category 34

Highest Avg Resolution Hours
1,326.32

Lowest SLA Category (100+ Incidents)
Category 55



Key Findings

Category 42 generated the highest incident volume with **3,138 incidents** while maintaining **79.51% SLA compliance**.

Category 46 combined high volume with weak performance, recording **2,356 incidents**, **316.02 average resolution hours**, and **47.54% SLA compliance**.

Category 34 recorded the longest average resolution time at **1,326.32 hours**.

Category 55 had the lowest SLA compliance among categories with at least **100 incidents**.

Category 61 recorded the highest reopen rate among categories with at least **100 incidents** at **2.35%**.

Assignment Group Performance

Comparing workload, resolution time, SLA compliance, and reopen rates across support groups

Highest Volume Group
Group 70

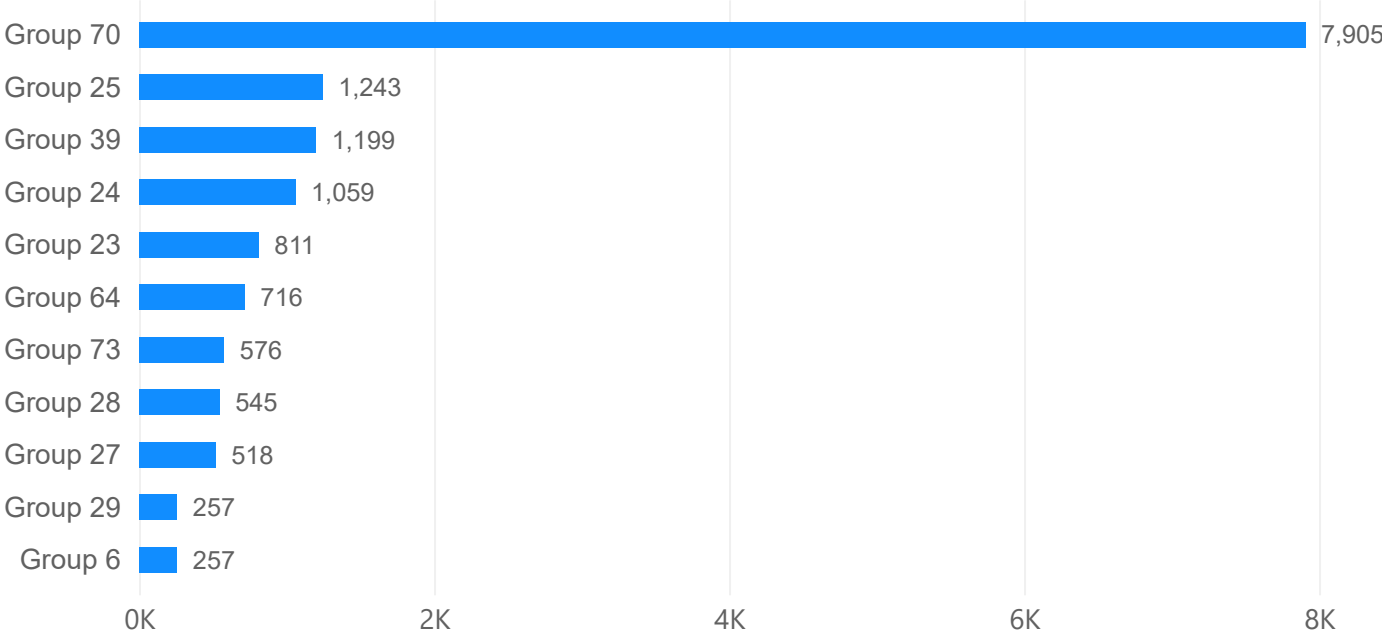
Highest Incident Volume
7,905

Highest SLA Group
Group 64

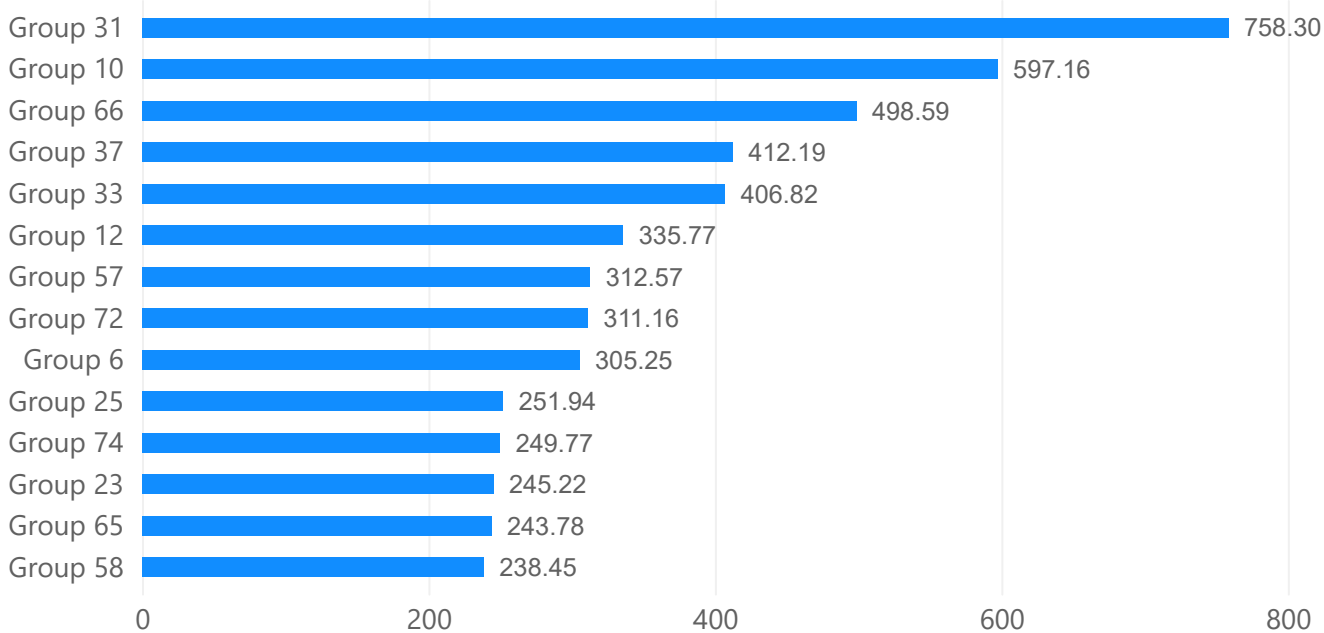
Highest SLA Compliance %
89.11%

Lowest SLA Group
Group 12

Top Assignment Groups by Volume



Average Resolution Hours by Assignment Group



Key Findings

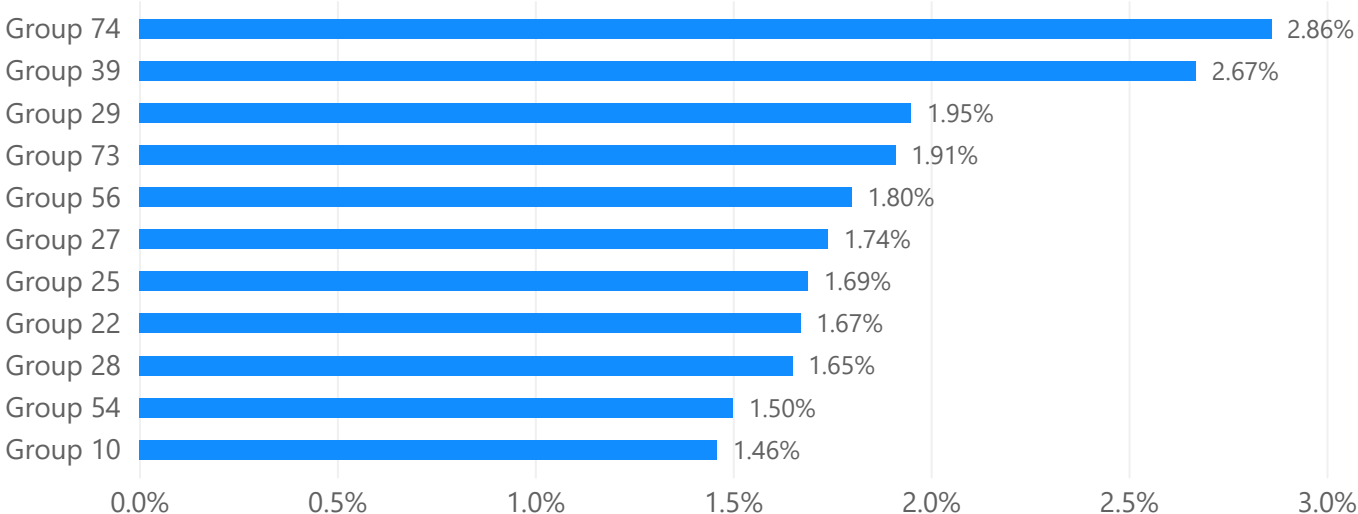
Group 70 handled the highest incident volume with **7,905 incidents** while maintaining **82.88% SLA compliance**.

Group 64 recorded **89.11% SLA compliance** and averaged only **18.75 hours to resolution**.

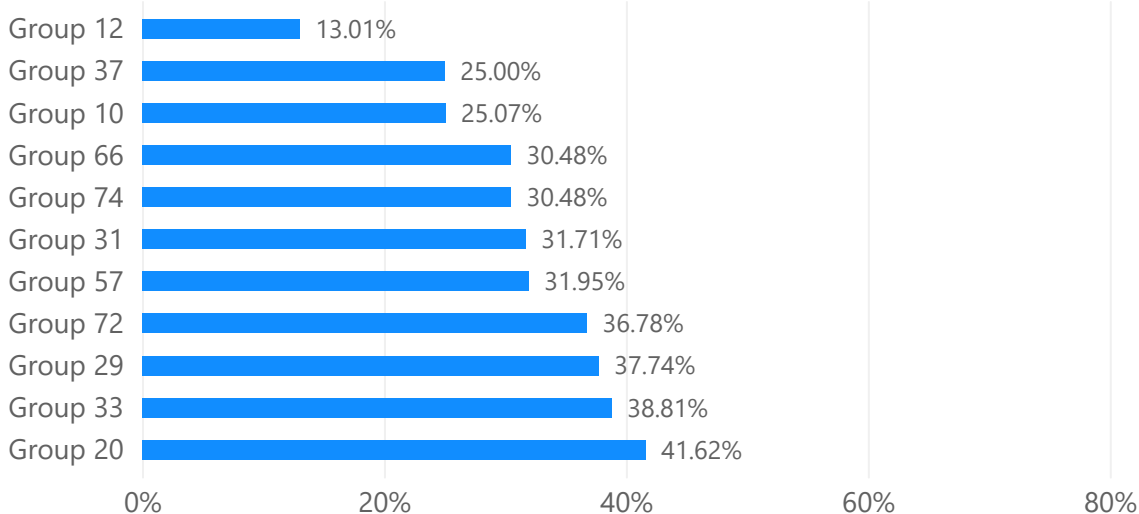
Group 12 recorded the lowest SLA compliance among groups with at least **100 incidents** at **13.01%**.

Group 10 averaged **597.16 resolution hours** with only **25.07% SLA compliance**.

Reopen Rate by Assignment Group



SLA Compliance % by Assignment Group



Priority & Reopen Analysis

Examining how incident priority and reopen activity relate to resolution performance and SLA compliance

Highest Volume Priority
Priority 3

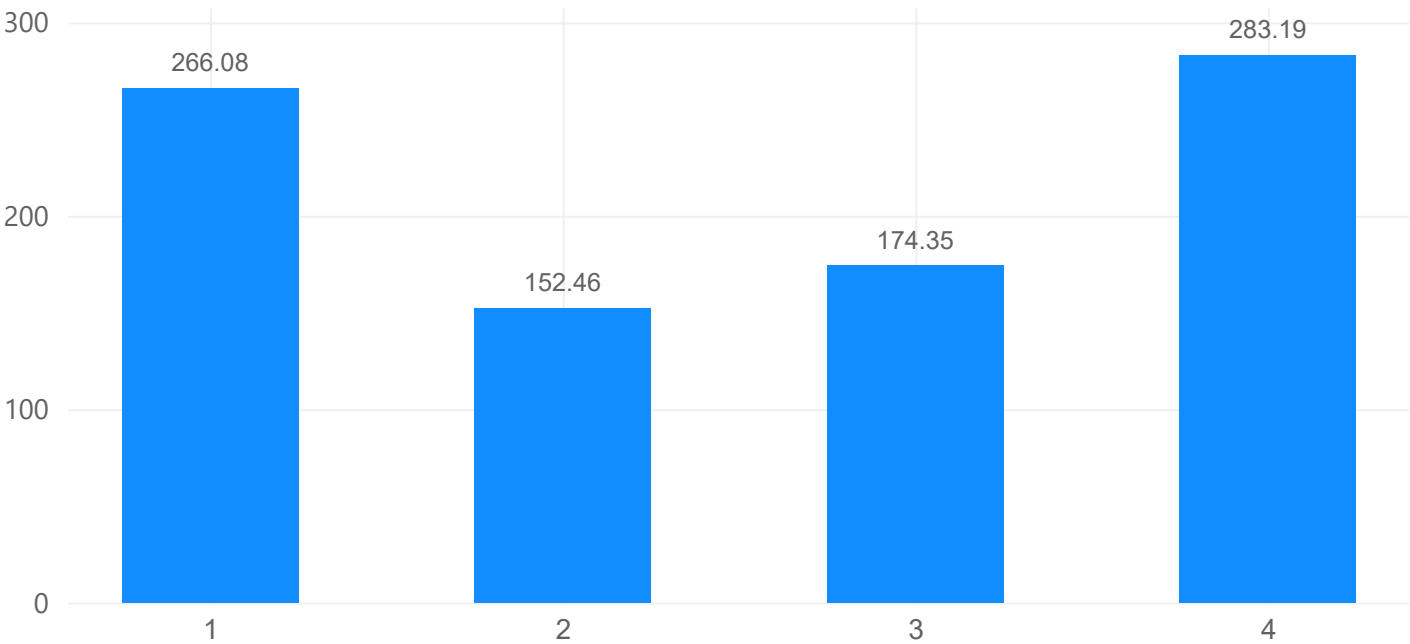
Highest Incident Volume
22,010

Reopened Incidents
274

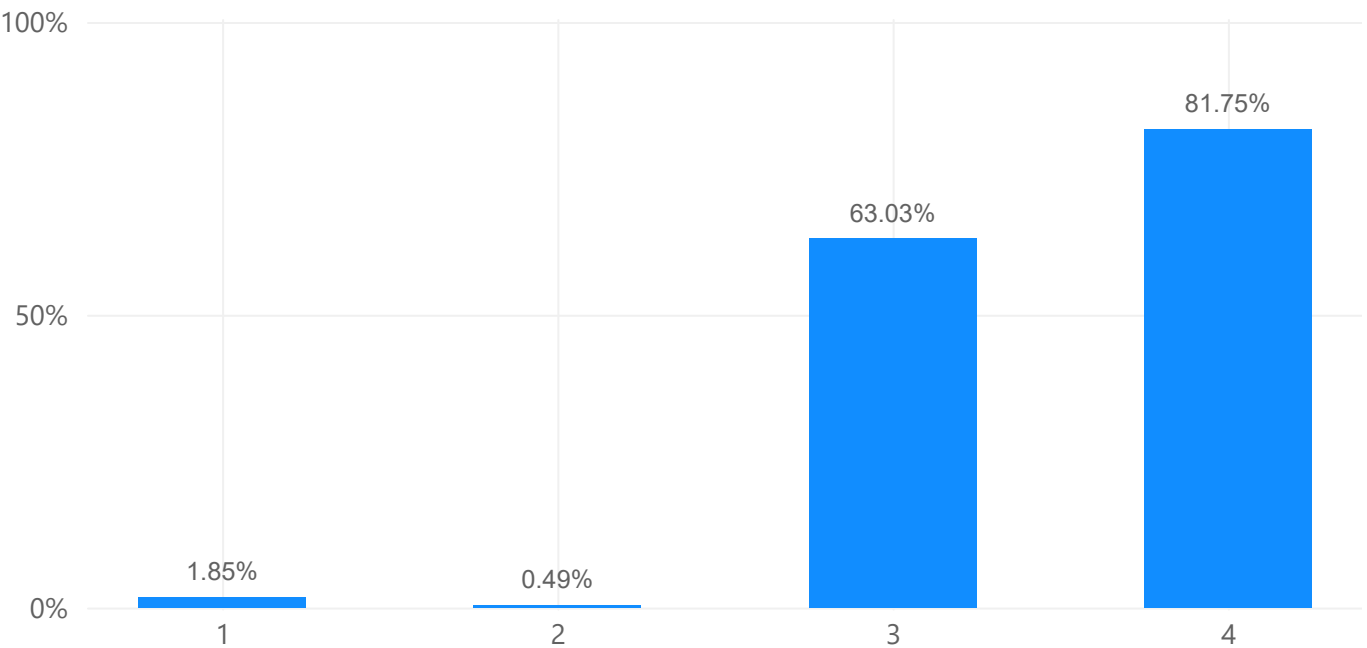
Reopened Avg Resolution Hours
457.37

Reopened SLA Compliance %
24.09%

Average Resolution Hours by Priority



SLA Compliance % by Priority



Key Findings

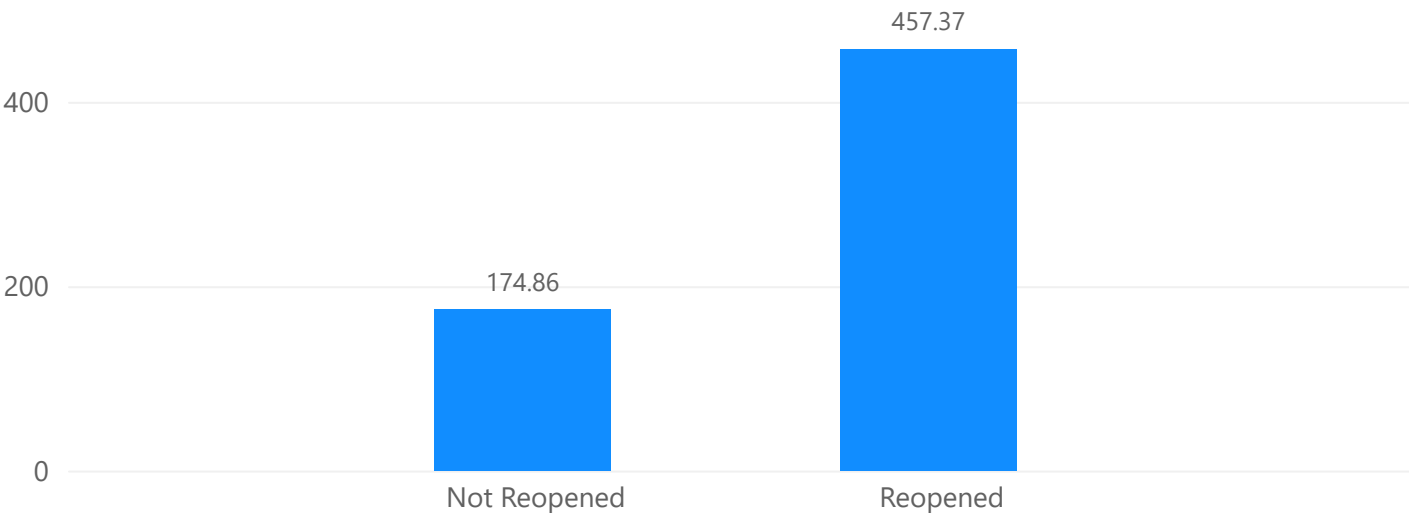
Priority 3 represented most of the analyzed workload with **22,010 incidents**.

Priority 1 recorded the longest median resolution time at **80.24 hours**.

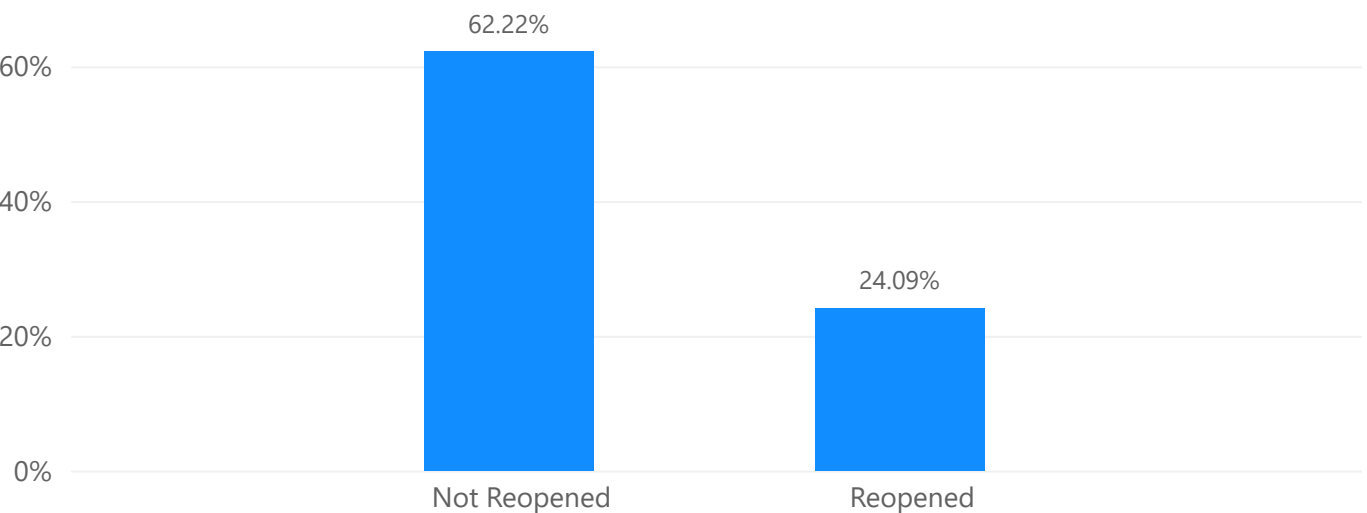
Reopened incidents averaged **457.37 resolution hours**, compared with **174.86 hours** for **not reopened incidents**.

SLA compliance declined from **62.22%** for **not reopened incidents** to **24.09%** for **reopened incidents**.

Average Resolution Hours by Reopen Status



SLA Compliance % by Reopen Status



Monthly Performance Trends

Tracking incident volume, resolution performance, SLA compliance, and reassignment activity over time

Peak Incident Month
March 2016

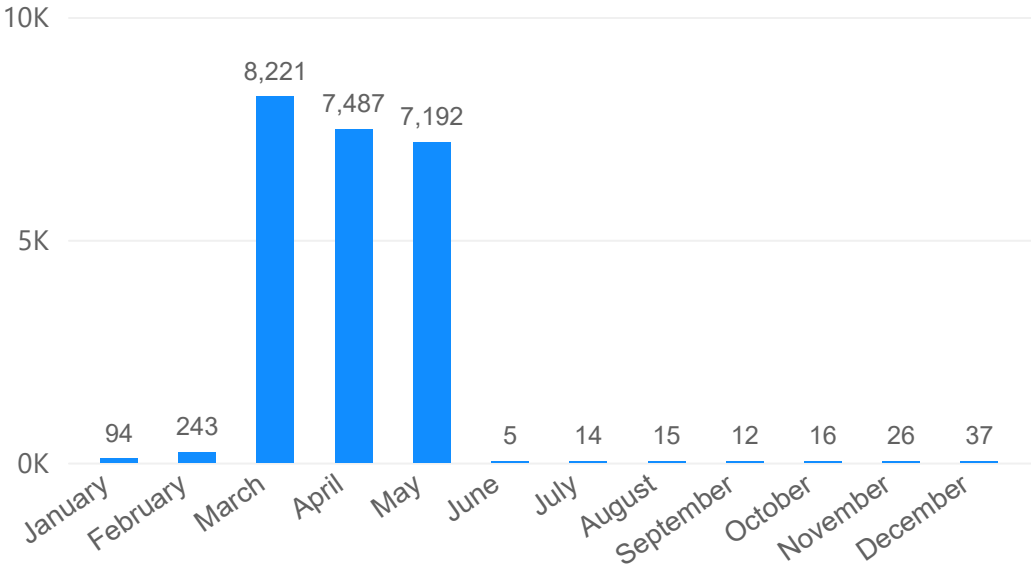
Peak Monthly Volume
8,221

Lowest SLA Month
March 2016

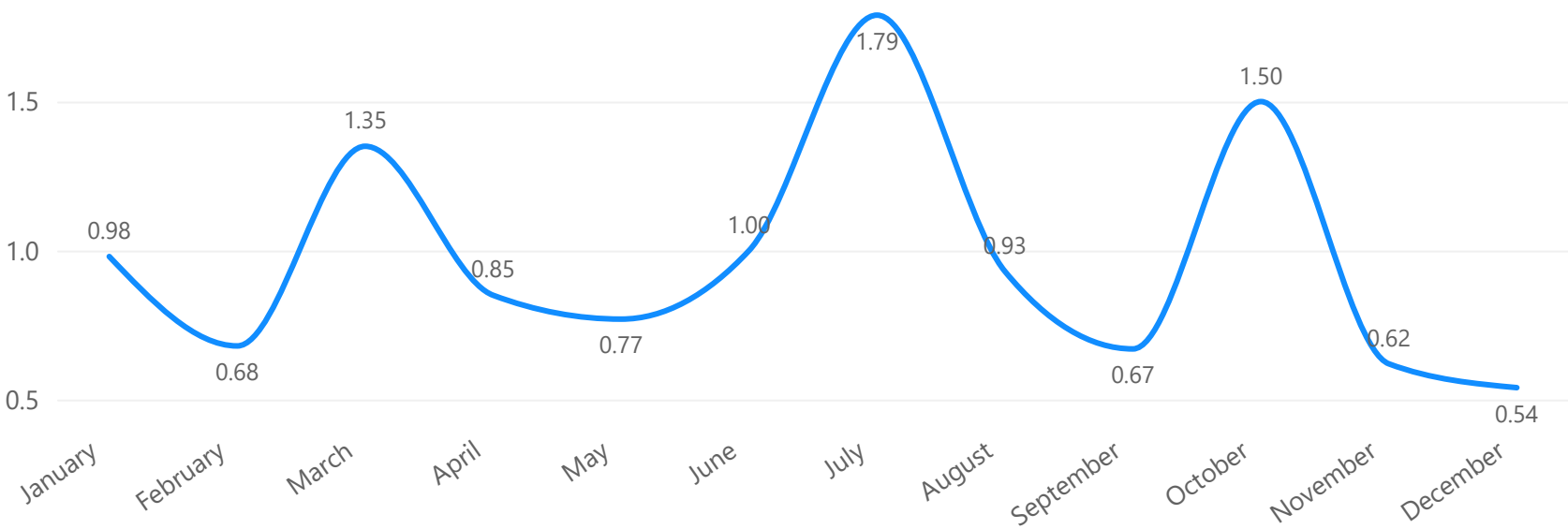
Lowest SLA Compliance
35.94%

Best SLA Month
May 2016

Incident Volume by Month



Average Reassignments by Month



Key Findings

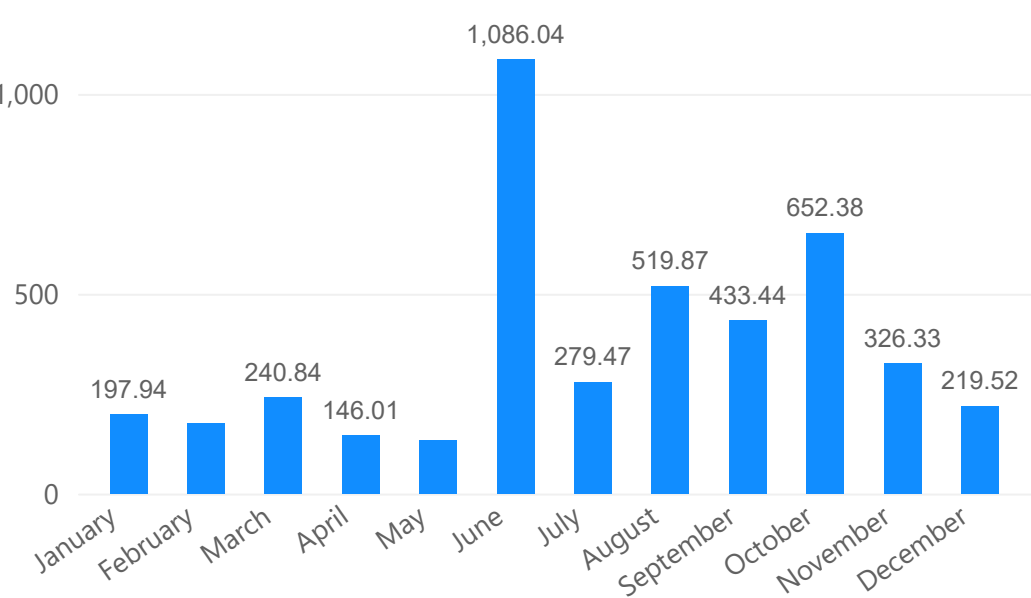
March 2016 recorded the highest incident volume with **8,221 incidents** and the lowest SLA compliance among major volume months at **35.94%**.

SLA compliance improved to **75.76% in April** and **76.63% in May**, even though both months recorded more than **7,000 incidents**.

Average resolution time decreased from **240.84 hours in March** to **132.76 hours in May**.

Later months contained substantially fewer incidents, so their performance results should be interpreted cautiously.

Average Resolution Hours by Month



SLA Compliance by Month



Data and Methodology

Dataset documentation, metadata, source attributes, analytical structure, and project limitations

Dataset Overview

"This event log was extracted from data gathered from the audit system of an instance of the ServiceNow platform used by an IT company and enriched with data loaded from a relational database."

The Incident Management Process Enriched Event Log contains 141,712 event records representing 24,918 unique incidents. The dataset was extracted from an anonymized ServiceNow environment and includes 36 attributes describing incident lifecycle activity, assignment history, service categorization, priority management, ownership, and SLA performance.

Important Limitations

- Categories and assignment groups are anonymized.
- Missing values represent unknown information.
- Resolution times are influenced by long-running incidents.
- Later reporting months contain substantially fewer incidents.
- Relationships identified in the analysis do not prove causation.

Dataset Citation

Amaral, C., Fantinato, M., & Peres, S. (2018). Incident Management Process Enriched Event Log. UCI Machine Learning Repository. DOI: 10.24432/C57S4H

Key Source Attributes

Incident Identification

- number
- incident_state
- active

Lifecycle & Timing

- opened_at
- resolved_at
- closed_at

Performance Metrics

- reassignment_count
- reopen_count
- sys_mod_count
- made_sla

Service Context

- category
- subcategory
- priority
- assignment_group

Process Ownership

- assigned_to
- opened_by
- resolved_by

Attributes Used In This Analysis

Resolution Analysis

- opened_at
- resolved_at
- closed_at

SLA Analysis

- made_sla

Reassignment Analysis

- reassignment_count

Reopen Analysis

- reopen_count

Category Analysis

- category

Assignment Group Analysis

- assignment_group

Priority Analysis

- priority

Trend Analysis

- opened_at

Dashboard Page	Primary Views
Reassignment Analysis	vw_Reassignment_Impact
Priority & Reopen Analysis	vw_Priority_Performance, vw_Reopen_Performance
Monthly Performance Trends	vw_Monthly_Performance
Executive Overview	vw_Incident_Summary, vw_Reassignment_Impact, vw_Category_Performance
Category Performance	vw_Category_Performance
Assignment Group Performance	vw_Assignment_Group_Performance

Metadata	Value
Event Records	141712
Unique Incidents	24918
Original Attributes	36
Privacy Status	Anonymized
Subject Area	Business
Data Structure	Multivariate & Sequential
Source Platform	ServiceNow
Repository	UCI Machine Learning Repository
Missing Values	Yes

About This Project | [GitHub link for my Portfolio](#)

Created by [Jake Sheeler](#) using Python, SQL Server, and Power BI to analyze 24,918 incidents from an anonymized ServiceNow environment.

Reporting Period: February 2016 – February 2017

Project Completed: October 7, 2026